Wipro — Aptitude Test 3

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Question 51. Two equal amounts of money are lent out at 6% and 5 % simple interest respectively at the same time. The former is recovered two years earlier than the latter and the amount so recovered in each case is Rs. 2800. Determine the amount that is lent out?

Question 52. The price of Maruti car has risen by 25% and the sales have come down by 4%. What is the total percentage change in revenue.

Solution:
Assume that the price of one car
= Rs. 100
Assume that the initial sales numbers is
= 100 cars.
Total revenue in the initial stage
= 100 x 100 = Rs. 10000
Current price of Maruti car = Rs. 125
Current sales number = 96
Current revenue = 96 x 125 = Rs. 12000
The percentage increase in revenue
= [(12000 – 10000)/10000]100
= 20% increase.

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Question 53. My friend collects antique stamps. She purchased 2 but found that she need to raise money urgently. So she sold them for Rs. 8000 each. On one she made 20% profit and on the other she

lost 20%. How much did she gain or lost
in the entire transaction?

Solution:
Selling price of each stamp = Rs. 8000.
Selling price of first stamp
= (120/100)CP = 8000
Cost price of first stamp
= 8000 x 100/120 = Rs. 6666.67
Selling price of second stamp

= (80/100)CP = 8000
Cost price of second stamp
= 8000 x 100/80 = Rs. 10000
Total Cost price = 16666.67
Total selling price = 16000
Loss percentage = 4%.
Shortcut:
(202/100) % loss ; (a2/100)% loss

Question 54. The total age of Old and Young is 48. Old was twice as old as Young when Old was half as old as Young will be when Young is three times as Old was when Old was three times as old as Young. How old is Old?

Solution:
When Old was three times as old as
Young
O = 3Y
Young is three times as old as Old, when
Old was trice as old as Young.
When Y = 3O when O = 3Y
So, Y = 9xO.
Now, Old is half as old as Young.
O = 9xO/2
And O = 18xO
Old = 30 and
Young = 18

Question 55. A father's age was 5 times his son's age 5 years ago and will be 3 times son's age after 2 years, the ratio of their present ages is equal to:

Solution:
Assume that the present age of father
and son is F and S.
(F – 5) : (S – 5) = 5 : 1 and
(F + 2) : (S + 2) = 3 : 1

From the ratios we get,
F – 5 = 5S – 25
F – 5S = - 20
---(1)
F + 2 = 3S + 6
F – 3S = 4
---(2)
By solving (1) and (2), we get
S = 12 and F = 40
Ratio of present ages of father and son
= 40 : 12 = 10 : 3

Question 56. If the circumference of a circle is 200 units, then what will the length of the arc

described by an angle of 20 degree?

Solution:
For 360o of the circle, the circumference
= 200 units.
For 20o of the circle we will get an arc.
If the length of the arc is x, then
20/360 = x/200
x = 4000/360
x = 11.11 units.

Question 57. A rectangle has twice the area of a square. The length of the rectangle is 14 cm greater than that side of the square whereas breadth is equal to side of the square. Find the perimeter of the square?

Solution:
Since breadth of the rectangle is same as
the side of the square and area of
rectangle is twice that of area of square,
the length of rectangle is equal to twice
the side of square.
Length of rectangle = 14 x 2 = 28
Breadth of rectangle = 14
Perimeter = 2(14+28) = 84cm.

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Question 58. If the area of a square increases by 69% then the side of the square will increase by what percentage?

Solution:
Assume that the are of square was 100
Now the area is 169.
Side of increased square = 13
Side of initial square = 10
Percentage increase of side = 30%

men can it feed?

Question 59. If the radius of a circle is decreased by 10% then its area is decreased by what percentage?

Solution:
Quantity of food = Number of men x
Number of days x Food per head
After

7
days
the
quantity
of
food
remaining = 3300 x 25 x 850
The same quantity food is provided for
the extra men.
Let us assume that the new number of
men added = N
The quantity of food for the new total
number of men
= (3300+N) x (25-8)x825
In
each
case,
the
quantity
of
food
remains the same.
So,
3300x25x850= (3300+N)x 17 x 825
N = 1700
The new number of men arrived = 1700

Solution:
Assume that the radius of circle initially
= 10 units
Area of the circle initially
= π(10x10) = 100 π
Current radius of the circle
= 9 units
Current area = π(9x9) = 81π
Percentage change in area
= 19% decrease.

Question 60. A circle has 2 parallel chords one of length 6 cm and other of length 8 cm. If the chords are in the same side of the center then distance between them 1 cm, find diameter of the circle.

Question 61. A garrison of 3300 men has provisions for 32 days, when given at a rate of 850 grams per head. At the end of 7 days a reinforcement arrives and it was found that now the provisions will

last 8 days less, when given at the rate of 825 grams per head. How, many more

Solution:
Work done by A on the first day = 1/9
Work done by B on the second day
= 1/12

Total work done in two days

1

1

12
=
+
7

9

36

The time taken to complete 35/36th of the work is = 10 days.
On the 11th day when A has to work, the remaining work = 1 – (35/36)

= 1/36
Time taken for A to complete 1/36th of the work = (1/36) x 9 = ¼ days.
Total time taken = 10.25 days

Question 62. On a certain pasture the grass grows at an even rate. It is known that 40 cows can graze on it for 40 days before the grass is exhausted, but 30 cows can graze there as long as 60 days. How many days would the pasture last if 20 cows were allowed to graze on it?

Solution:

r

x

4
1

Solution:
When the number of cows in decreased by 1/4th (from 40 to 30), the number of days increased by 50%.
When the number of cows is decreased by 1/3rd (from 30 to 20) the number of days will increase by 100%.
The number of days taken by 20 cows to graze the field = 120 days.

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r2 = 32 + (1 + x)2

r2 = 42 + x2

9 + (1+x)2 = 16 + x2

1 + x2 + 2x = x2 + 16 – 9
2x = 6 ■x = 3
r2 = 32 + (1 + 3)2

Question 63. A and B working separately can do a piece of work in 9 and 12 days respectively. If they work for a day alternatively with A beginning the work, in how many days the work will be completed?

r = 5
Radius of the circle is 5cm.
Diameter of the circle = 10cm.

Question 64. An empty tank be filled with an inlet pipe A in 42 minutes. After 12 minutes an outlet pipe B is opened which can empty the tank in 30 minutes. After 6 minutes another inlet pipe C opened into the same tank, which can fill the tank in 35 minutes and the tank is filled. Find the time taken to fill the tank?

Solution:
Let N be the time taken to fill the tank.
Pipe A is opened for N minutes.
Pipe B is opened for N – 12 minutes.
Pipe C is opened for N – 18 Minutes.
Fraction of tank filled by pipe A in 1 minute = 1/42
Fraction of tank emptied by pipe B in 1 minute = 1/30
Fraction of tank filled by pipe C in 1 minute = 1/35
So,

30
=
-
1
N – 18

N – 12

N

35
+

42

By solving the above, we get
5N – 7N + 84 + 6N – 108 = 210
4N = 234
N = 58.5 minutes

Question 65. A is 6 times as fast as B and takes 100 days less to complete a work than B. Find the total number of days taken by A and B to complete the work together.

Solution:
Let us assume that the one day work of B
= 1/B

Then one day work of A = 6/B
Time taken by B to complete the work
= B days.
Time taken by A to complete the work
= B/6 days.
Given, B – (B/6) = 100
5B/6 = 100
B = 120 days
If B can do it in 120 days, time taken for
A to complete the work = 120/6
= 20 days.
Time taken by both of them together
= [20 x 120/(20 + 120)]
= 2400/140
= 120/7 days.

Question 66. 30 men take 20 days to complete a job working 9 hours a day. How many hours a day should 40 men work to complete the job in 40 days?

Solution:
W1R2D2H2= W2R1D1H1
W = work done
R = resource or human power
D = number of days
H = hours per day
From the question, W1 = W2
So,
40 x 40 x H2 = 30 x 20 x 9
H2 = 3.375
= 3 hours 22 minutes 30 seconds

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Question 67. A, B and C can do a piece of work in 30, 45 and 90 days. How many days A alone do the work if he is assisted by B

and C on every 4th day?

Solution:
Work done by A in first three days
= 3/30 = 1/10
Work done on 4th day
= (1/30) + (1/45) + (1/90)
= 1/15
Total work done in every 4 days
= (1/10) + (1/15) = 1/6
In every 4 days 1/6th of the work I done.
Time taken to complete the whole work
= 6 x 4 = 24 days.
Days worked by A alone = 24 – 6 = 18

y = 1/60 = 1/C
One child can do the work in 60 days.
x = 1/15 = 1/M
One man can complete the work in 15 days.

Question 68. 16 men can complete a work in 24 days while 48 children can do it in 16 days. 12 men started the work , after 14 days 12 children joined them. In how many days will they do the remaining work?

Question 69. Two men and 7 children complete a certain piece of work in 4 days while 4 men and 4 children complete the same work in only 3 days. The number of days required by 1 man to complete the work is?

W i p r o

Solution:
Assume that 1 man can complete the work in M days and 1 child can complete the work in C days.
Work done by one man in 1 day = 1/M
Work done by 1 child in one day = 1/C
Therefore, from the question:
(2/M) + (7/C) = 1/4 and
(4/M) + (4/C) = 1/3
Assume (1/M) = x and (1/C) = y
Then,
2x + 7y = 1/4
4x + 4y = 1/3
By solving the above equations, we get

Question 70. If 5 men take an hour to dig a hole, then how long would 12 men take to dig to hole of the same type?

Solution:
If 16 men can complete the work in 24 days, time taken for 12 men to complete the work will be:
(16/12) = (x/24)
x = 32 days.
If 48 children can complete the work in 12 days, time taken for 12 children to complete the work will be:
(48/12) = (y/16)
y = 64 days.
Time taken by 12 men and 12 children to complete the work
= [32x64/(32+64)] = 64/3 days.
Work done by 12 men in 14 days
= 14(1/32) = 7/16
Remaining work after 14 days = 9/16
Time taken for 12 men and 12 children to complete 9/16th of the work
= (9/16)(64/3)
= 12 days.

Solution:
Time and resource are inversely proportional.
So,
If x is the time taken by 12 men to dig the hole, then
5/12 = x/1
x = 5/12 hours
= 25 minutes.
12 men will dig the hole in 25 minutes.

Question 71. A pipe can be fill a tank in 20 min but there is a leakage in it which can empty the full tube in 60 min. In how many minutes the tank can be filled if both the pipe and the leakage are opened?

Solution:
Fraction of tank filled in 1 minute by the inlet pipe = 1/20
Fraction of tank emptied by the leakage in one minute = 1/60
Fraction of tank filled if both are opened together = (1/20) – (1/60)
= 1/30
Time taken to fill the tank = 30 minutes.

Question 72. A solar powered car is being test driven. The vehicle is driven at 30 mph under solar power and 40 mph under regular power. The trip to the nearest

town takes 45 min using both solar and regular power whereas the return trip takes 50 min using only solar power. On the trip to the town find the distance driven using regular power?

Solution:
Using the return travel we can find the distance of the nearest town

Distance of the nearest town
= Speed under solar power x time
= 30 x (50/60)
= 25 miles.
Let us assume that in the onward journey, the vehicle is driven under regular power for T minutes and under solar power for (45 – T) minutes.
Distance travelled by regular power
= (40/60) x T
Distance travelled by solar power
= 30 x (45 – T)/60
Both distances add up to 25 miles.
So,
(40/60)T + 30 x (45 – T)/60 = 25
40T + 1350 – 30T = 1500
10T = 150
T = 15 minutes.
Distance travelled by regular power
= (15/60) x 40
= 10 miles.

Question 73. The ratio of time taken to run by Harish and Dev is 4:3 and thus Dev wins by 360m. What is the distance of the race course?

Solution:
Let us assume that the time taken by Harish and Dev are 4 seconds and 3 seconds.
Harish has to cover 360m in 1 second to finish the race.
Speed of Harish = 360m per second.
He runs for 4 seconds to finish the race.
Length of the race track = 4 x 360
= 1440m

Question 74. A passenger train takes two hours less for a journey of 300km if its speed is increased by 5 kmph from its normal speed. The normal speed is.

Solution:
Assume that the actual speed = S and actual time taken = T

Then,
T = 300/S and
T – 2 = 300/(S+5)
T = [300/(S-5)] + 2

300/S = [300/(S+5)]+2
300S + 1500 = 300S + 2S2 + 10S
2S2 + 10S – 1500 = 0
S2 + 5S – 750 = 0
By solving this, we get S = 25
Speed of the train is 25kmph.

Question 75. A man travels by bus for 20 hours and they by train for 5 hours. If the average speed of the bus was 20 kmph and that of the entire journey was 24 kmph what was the average speed of the train?

Solution:
Total time taken for the journey
= 20+5 = 25 hours.
Total distance covered
= Time x Average speed
= 25 x 24 = 600km.
Distance travelled by bus
= 20 x 20 = 400km.
Distance travelled by train
= 600 – 400 = 200km.
Average speed of train
= 200/5 = 40kmph.